PURPOSE: Anchoring exercise intensity with ratings of perceived exertion (RPE) can be used to examine the mechanisms underlying the perception of effort during fatiguing tasks. The purpose of the present study was to examine the fatigue-related patterns of responses for force and electromyographic amplitude (EMG AMP) during sustained isometric, leg extension muscle actions anchored by RPE.
METHODS: Ten recreationally active men (mean ± SD: 22.9 ± 2.7 yr) performed 3 randomly ordered, sustained submaximal isometric leg extension muscle actions (knee joint = 120°) anchored at RPE values of 2, 5, and 8 (OMNI-RES 10-point scale) to volitional exhaustion or a maximal time limit of 5-min. Maximal voluntary isometric contractions (MVIC) were performed prior to and immediately following each sustained isometric muscle action and bipolar surface EMG signals were recorded from the vastus lateralis muscle during the sustained muscle actions. Linear regression analyses were used to examine the force and EMG AMP vs. time relationships during each sustained isometric muscle action.
RESULTS: The pretest MVIC values (62.1 ± 14.4, 62.6 ± 14.7, and 63.5 ± 12.7 kg) were highly reliable (ICC= 0.90) and were not significantly ($p=0.930$) different. The posttest MVICs were significantly ($p<0.001$) less than pretest for each sustained isometric muscle action. There were significant ($p<0.001$) differences between all the mean times for the sustained muscle actions (RPE 2 = 300.0 ± 0.0 s; RPE 5 = 202.0 ± 95.5 s; RPE 8 = 72.7 ± 27.6 s). Furthermore, the percent decline in force decreased ($p<0.001$) with each increase in RPE (percent decline for RPE 2 = 61.0 ± 18.5%; RPE 5 = 47.4 ± 19.6%; RPE 8 = 24.9 ± 13.2%). For normalized EMG AMP, there were significant negative, linear relationships for RPE=2 ($p=0.006$) and RPE=5 ($p=0.003$) vs. time of the sustained isometric leg extension.
CONCLUSION: For the RPE=2 and RPE=5 trials, EMG AMP and force decreased and, therefore, were dissociated from RPE. At RPE=8, however, EMG AMP was unchanged, but force decreased. Thus, EMG AMP tracked RPE, but was dissociated from force. Together, these findings suggested that the perception of effort during sustained, submaximal isometric leg extension muscle actions were not mediated by fatigue-related changes in force or EMG AMP.

---

197 Board #38 May 30 9:30 AM - 11:00 AM

A Comparison of Muscle Unit Activation during Biceps Curl Exercise at 40, 60, 80 and 100% of 1 Repetition Maximum

Alex Zykoff, Mike Aquino, Jacqueline Pellechia, John Petrizzo, John Wygand, FACSM, Robert M. Otto, FACSM. *Adelphi University, Garden City, NY.* (Sponsor: Robert M. Otto, FACSM)

(No relevant relationships reported)

The optimal intensity of resistance training exercise is subject to ongoing debate. There is universal agreement that electromyography (EMG) is an excellent index of motor unit (MU) activation or recruitment. The amount of EMG activity can infer the relative intensity of the exercise as compared to a maximal voluntary contraction (MVC) of a specific muscle, performing a specific exercise within one specific individual.
PURPOSE: To determine the magnitude of muscle fiber recruitment (activation) of the Bicep Brachii during various loading of the muscle (40, 60, 80, and 100% of 1 RM) during bicep curl (BC) exercise performed to momentary muscular fatigue (MMF), 12 subjects (age 22.5±1.0 yr, ht. 169.4 cm±11.1 cm, body mass 75.3 kg±17.5 kg, 7♀) with resistance training experience, volunteered.
METHODS: A familiarization trial on a selectorized BC machine was performed and the 1 RM was titrated for each subject. In addition, a maximal voluntary contraction (MVC) was obtained for the BC exercise against an immovable load on the same selectorized machine. MU recruitment was estimated by EMG. All subjects adhered to a 3-1-3-1 sec repetition duration during all trials. All trials were randomly assigned and terminated at MMF.
RESULTS: Statistical analysis by ANOVA ($p<.05$) with repeated measures was applied to these data. Percent MVC for the concentric portion of the final repetition were 62.5±27.8, 71.7±24.0, 72.8±18.4, and 79.6 ±18.0 at 40, 60, 80, and 100% 1RM, respectively, with NSD among 60, 80 and 100%1RM. With the exception of the 40% trial, all trials revealed similar peak MU recruitment, despite different loading. The recruitment of ~70% of MVC at MMF may, in part, be attributed to the level of resistance training of the subject's and their ability to activate MU's.
CONCLUSION: The size principle of MU recruitment is supported, in that peak recruitment of motor units can be accomplished at any workload greater than 60% 1 RM**,** provided MMF is achieved. Intensity is best defined by the final repetition, as opposed to the initial repetition (i.e. % of 1RM).

---

198 Board #39 May 30 9:30 AM - 11:00 AM

The Effect Of Using A Weightlifting Belt On Muscle Activation During A Conventional Deadlift

Jacqueline Pellechia, John Petrizzo, Robert M. Otto, FACSM, John Wygand, FACSM. *Adelphi University, Garden City, NY.* (Sponsor: John Wygand, FACSM)

(No relevant relationships reported)

The use of weightlifting belts for free weight training and competition is commonplace. Theoretically, the use of the belt in conjunction with a simultaneous valsalva maneuver provides additional support to the spine by facilitating increased stabilizing pressure inside the abdominal cavity. This is accomplished by isometric contraction of the core muscles in the abdominal wall and lower back with support from the belt. Generally lifts with 5-15% greater loading are reported when wearing a belt**.**
PURPOSE: The purpose of this study was to determine the magnitude of motor unit activation in the biceps femoris (BF), gastrocnemius (G), and trapezius (T) during a conventional deadlift (DL) wearing a weightlifting belt (B) or without a belt (NB).
METHODS: 10 subjects with resistance training experience (age 22.2 ± 0.78, ht. 170.43 ± 11.19 cm, and body mass 81.72 ± 21.69 kg, 6♀) volunteered to perform a familiarization trial of the conventional DL technique of three repetitions at 70% of their self-reported one repetition maximum (1RM). A maximal voluntary contraction (MVC) was obtained while subjects exerted a maximal force against an immovable load at approximately 35% DL ROM. After safe lifting form was confirmed, a 1RM was titrated and randomized trials of B and NB were performed to momentary muscular fatigue (MMF) at 80% of subject's 1RM with a minimum of 30 minutes between trials. MU recruitment was estimated by EMG. All subjects adhered to a 3-1-3-1 sec repetition duration and terminated the lift at MMF on the concentric phase.
RESULTS: Statistical analysis by dependent t test was applied to these data and revealed no significant difference($p>.05$) between trials with motor unit activation for BF of 98.0% and 93.7%, for G of 92.2% and 80.7%, and for T of 72.3% and 71.0%, for B and NB trials, respectively.
CONCLUSION: The use of a properly aligned and fitted weight- lifting belt did not yield a significant difference in muscle activation between the (B) and (NB) trials for the tested muscle groups while performing the conventional DL. Thus, a weight lifting belt may augment the lifter's perception, but it does not enhance or detract from motor unit activation.

---

199 Board #40 May 30 9:30 AM - 11:00 AM

Comparison of EMG Muscle Activity and Turn Times Using Orthosis and Stock Insoles During Skiing

Heidi Nunnikhoven. *Montana State University, Bozeman, MT.*

(No relevant relationships reported)

New alpine ski boots are generally sold with generic, foam rubber stock insoles which may lead to over-pronation of the foot due to a lack of subtalar support. The use of stock insoles may result in inefficient movements of the foot and reduced transfer of force throughout the turn cycle. A flexible orthosis insole is designed to provide support of the foot and while still allowing for the rotation of subtalar joints which may improve the transfer of force from the foot to the boot to the ski edge.
PURPOSE: The aim was to compare the EMG activity of the tibialis anterior (TA), peroneus longus (PL), and gastrocnemius (GA) muscles and turn times when skiing using heat molded, flexible orthosis insoles (OI) and stock insoles (SI).
METHODS: Following IRB approval, five elite skiers (including 3 former national champions) skied two runs on a 13 brush-gate course with a 22° pitch. Each skier was asked to ski consistently for two trials, one trial with OI and one trial with SI. The SI were specific to the given boot company of the skier. Brush gates were set at a diagonal distance of 12 m. Surface EMG activity was processed by RMS. Turn times were assessed with a motion sensor attached to the ski boot. Data analysis utilized a 1-way ANOVA for comparison of EMG activity and time per turn.
RESULTS: EMG activity for the TA and PL were significantly lower in the OI trial compared to the SI trial: 3.80 ± 1.28 mV vs. 4.26 ± 1.23 mV ($p=.04$) and 2.50 ± 0.59 mV vs. 2.80 ± 1.29 mV ($p=.01$), respectively. There was no statistical difference between insoles for GA activity: 1.81 ± 0.38 mV vs. 2.01 ± 0.7 mV ($p=.09$) for OI and SI. Average time per turn was significantly faster for OI than SI: 1.038 ± 0.012 sec vs. 1.058 ± 0.024 sec ($p=.03$).

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.